

Multi-scale latent diffusion using hierarchical diffusion models for interpretable music generation

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In recent years, significant advances in deep neural audio synthesis have led to the development of exciting new tools for composition and sound design. Most notably, recent progress of generative models, such as Generative Adversarial Networks (GANs) and Variational Auto-Encoders, have led to powerful neural audio codecs such as RAVE [3] or Encodec [5], which allow to project audio waveform to a low dimensional latent space capturing salient features of the data. However, it is not straightforward to use such models to generate music tracks with long-term coherency, as their receptive field remains limited to a temporal scope of a few seconds. Early work using autoregressive networks to predict temporal trajectories in the latent space has yielded impressive results, at the cost of expensive training and long synthesis time [7] [4] [2]. More recently, diffusion models have been used to synthesize high quality audio while remaining stable in training, and have been successfully applied to music generation [1] [6]. However, those approaches are mostly designed for text-to-music applications. Although such models exhibit impressive performance, they offer coarse, static control over the generation, which severely hamper their use in creative contexts. The DiffusionAutoencoder (DiffAE) [8] approach proposed to combine an encoder that capture high-level semantic information about the generation target with a stochastic diffusion decoder conditioned on the semantic representation that captures the finer variations. We argue that this model initially proposed for image generation does not fit well with the inherent multi-time scale nature of audio signals. The goal of this internship is to build upon DiffAE and design a temporal hierarchical generative model [9] [10] based on cascaded diffusion autoencoders, each one capturing semantic information at different levels of abstraction. Hence, the model would be able to generate long-term coherent signals by sampling the highest level semantic representation, while being able to generate variations of a sound by sampling from the lower temporal levels. The intern should derive the model formulation and architecture and validate the correlation of semantic content of each representation with interpretable musical features. Finally, the intern should study how control signals can be injected in the different representation levels of the model to enable creative musical applications.

References

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